Response to Reviewer 2

Manuscript **TP-1083** — *The Incommensurability Principle in Biological Transport*
Transport Phenomena (De Gruyter Brill)

We thank Reviewer 2 for a detailed and constructive review. The four points raised have led to substantive additions that make the theory's assumptions explicit, broaden and sharpen its empirical test, and improve its accessibility. Below we reproduce each comment (paraphrased from the report) and detail our response. The highlighted copy uses a distinct colour per source, as the editor requested: blue for Reviewer 1, red for Reviewer 2, green for editorial requests, and magenta for author-initiated refinements.

Comment 1 — Assumptions and alternatives.

Reviewer comment.

This manuscript introduces an original and mathematically rigorous theoretical framework to account for the conserved branching exponent in biological transport networks, constituting a substantive extension of the authors' previous work. Although the framework is mathematically sophisticated and internally consistent, several components would benefit from further clarification and empirical validation.

Several key derivations rest on assumptions—evolutionary optimisation, informational constraints, metabolic fitness landscapes—that, while biologically plausible, may not be uniquely warranted. A more extensive discussion of these assumptions and their possible alternatives would be valuable.

Response. We have added a dedicated closing subsection to the Discussion, "Underlying Assumptions and Alternative Hypotheses" (Section 7), that makes explicit the five load-bearing assumptions and, for each, states the principal alternative and—where one exists—the observation that would discriminate between them:

(A1) **Optimality**, vs. developmental canalisation, physical self-organisation, and neutral evolution. We explain that self-organisation is not in opposition to optimality—local shear-homeostatic remodelling (Hu & Cai; Ronellenfitsch & Katifori) converges to the same optimum, so it is the *mechanism* and the minimax the *target*—whereas the genetic-blueprint alternative is the explicit falsification target of the ontogenetic transition test.

(A2) **Minimax vs. average-case.** We acknowledge a genuine identifiability limitation: a broad-prior Bayesian (expected-cost) optimum converges to nearly the same α^*, so the two are hard to separate with present data. We adopt the minimax as the more conservative, assumption-light choice, requiring no commitment to a particular activity prior.

(A3) **Informational bound**, vs. long-range biological coordination or genetic pre-specification—the former constrained by the phase-lag blind spot of Section 2.

(A4) **Metabolic cost currency**, vs. other selective currencies (developmental robustness, fatigue life, thermoregulation, repair, reproduction). This is mitigated by the parameter-independence (Topological Rigidity) result, and multi-currency organs (e.g. the kidney) are shown to fall *outside* the two-channel minimax by construction.

(A5) **Incommensurability itself**, vs. a single unified energy budget; its empirical signature is the dual-attractor structure.

A closing "Epistemic status" paragraph states plainly which assumptions are falsifiable, which are supported, which delimit the domain, and which (A2) cannot currently be separated from its leading alternative. *Location:* Section 7.

Comment 2 — Broader empirical validation.

Reviewer comment.

Strengthen and broaden the empirical tests of the Incommensurability Principle and of the minimax attractor across a wider diversity of biological systems (beyond the present set of networks).

Response. We agree that empirical scope is central. We have therefore strengthened the empirical framing of the existing 27-network catalogue, making explicit that it constitutes a discriminating, pre-specified, cross-kingdom test of the principle's central prediction: that the realised exponent tracks *which* incommensurable cost channels are physically active. We added (Section 6, "Independent Datasets", with a twin framing paragraph in Supplemental Section S6) an explicit statement of the sorting rule:

- non-pulsatile networks—botanical xylem, airways, leaf venation, slime mold, mycelium, quasi-static venous return (ten systems, all observed in 2.80–3.15)—at the viscous attractor $\alpha_t \to 3.0$;
- pulsatile mammalian arteries at the minimax $\alpha^* \approx 2.7$;
- wave-dominated or planar ($d = 2$) networks near $\alpha_w \to 2.0$.

The sorting holds across three biological kingdoms (animal, plant, fungal/protist), both embedding dimensions, and pathological states—a span no single-attractor law (Murray's $\alpha = 3$ or area-preserving $\alpha = 2$ alone) can reproduce—and is falsifiable in two directions: a non-pulsatile network at $\alpha \approx 2.7$, or a high-Womersley pulsatile conduit at $\alpha \approx 3.0$, would refute it. We also identify the most informative directions for extension *beyond* the present set: taxa with divergent cardiac duty cycles (avian and reptilian circulations), invertebrate open circulatory systems, and engineered organ-on-chip networks with programmable pulsatility.

Comment 3 — Uniqueness of the linear cost functional (central point).

Reviewer comment.

The claim that the linear fractional excess is the unique *admissible cost functional must be supported more rigorously: explicitly consider alternative cost formulations, assess their limitations, and justify excluding them in favour of the linear form.*

Response. This is the central methodological point, and we have addressed it directly. Arguments that were previously scattered through the text (the exclusion of the logarithm, the weighted-Jensen consistency condition, and non-substitutability) are now consolidated into a single new Supplemental subsection, "Systematic Exclusion of Alternative Cost Functionals" (Section S7). It states four admissibility requirements,

(i) scale invariance,

(ii) compositional (weighted-Jensen) consistency,

(iii) Onsager linear response near the optimum,

(iv) non-substitutability of the two cost channels,

and tests six candidate functionals against them in a summary table: the linear $x - 1$, quadratic $(x-1)^2$, logarithmic $\ln x$, power-law $x^q - 1$, an absolute-threshold form, and a multiplicative $\mathcal{C}_{\text{wave}} \cdot \mathcal{C}_{\text{transport}}$. Only the linear excess satisfies all four; each alternative fails at least one (the quadratic, logarithmic, and power-law forms violate compositional consistency and Onsager linearity; the absolute-threshold form breaks scale invariance; the multiplicative form violates non-substitutability). We state explicitly that, within these four admissibility requirements, the linear excess is the only surviving form; relaxing any one of them is a concrete falsification target. A pointer to Section S7 was added at the non-substitutability remark in Section 3. *Location:* Section S7 and Section 3.

Comment 4 — Expository density.

Reviewer comment.

Several formal constructs in Sections 2–6 are introduced too quickly; expand the intermediate steps and add deeper biological interpretation of the formal results. Notwithstanding these concerns, the manuscript offers a significant contribution to the theoretical understanding of adaptive biological transport networks and their emergent scaling properties.

Response. We have carried out a systematic accessibility pass, working through the manuscript one section at a time from the beginning and inserting plain-language "bridge" passages wherever a formal construct was previously introduced too quickly. Each bridge either supplies a missing intermediate step or gives the biological/physical meaning of a formal result, and adds text without removing or altering the original derivations. The new passages are:

- **Section 1 (Introduction).** A word-level reading of the Kinematic Matching theorem (the Womersley number as a "switch": viscous resistors below Wo_c, pulsatile waveguides above it; two thresholds $\Rightarrow$ incommensurability), placed between the theorem statement and its proof; and a short gloss of what an *evanescent mode* is.
- **Section 2 (the no-go proposition).** A bricklayer analogy stating, in plain terms, what the proposition claims—that a purely local rule cannot keep the whole tree scale-free because it would need the non-local generation index—which also motivates Step 3 of the proof. (The intermediate derivation of the cost ratio $\Lambda(r) \propto r^2$ from the gradient scalings of Step 1 was likewise made explicit.)
- **Section 3 (gauge / ATP).** Three bridges: (i) reconciling the "absolute" and "fractional" cost statements via within- versus across-organism comparison; (ii) a plain reading of the Onsager-linearity theorem (linear in fractional waste, quadratic in geometric deviation); and (iii) a gloss of what the weighted-Jensen equation actually demands (boundary-independent scoring of subsystems).
- **Section 4 (architectural invariance).** A "game against Nature" framing of the minimax (the network picks the geometry, an adversary picks the worst operating weight, the saddle is the geometry immune to that choice), plus a one-line intuition for the envelope-theorem step.
- **Section 5 (single-mechanism limits).** A paragraph giving the physiological meaning of the two limits: the static limit as the regime of small and non-pulsatile vessels (Murray's law), the wave limit as large conduit arteries (area-preserving impedance matching), and the interior minimax as the mid-calibre pulsatile beds that morphometry actually samples.
- **Section 6 (architectural transition).** A plain-language account of the phase decoupling: because the wave responds to the square root of the fluid admittance, the fluid and the wave "switch on" at different body sizes, so they can never match simultaneously—the concrete meaning of incommensurability. (The intermediate algebra of the intermediate step $\Lambda(r) \propto r^2$ noted above complements this.)
- **Section 9 (retinal paradox).** An explicit up-front statement of the paradox (two-dimensional diameter scaling but non-planar bifurcation angles within one network) and of its resolution by mechanical–fluidic decoupling.

Apart from the new assumptions subsection described above (Comment 1), we did not add further expository bridge passages to Sections 7 and 8: as synthesis and conclusion they are already discursive and carry extensive interpretive remarks, so additional exposition there would be padding rather than clarification. In the Supplemental Material, whose detailed derivations are reference material, we added a single orienting sentence at the head of the densest sections—the admittance expansion (S1), the structural-residual decomposition (S3), and the thin-wall limit (S5)—stating what each establishes and why, without altering any derivation. *Location:* Sections 1–6 and 9 of the main text, and Supplemental Sections S1, S3, S5.

Closing. These changes leave the manuscript clearer and better bounded. Beyond the reviewer-driven changes (shown in red), the highlighted copy also marks a set of author-initiated refinements (magenta). These include a corrected citation (the Fåhræus–Lindqvist viscosity figure, re-stated to match the source), consistent disambiguation of previously overloaded symbols (η_d, the cardiac duty cycle, vs. η, the cost-mixing weight; and β, the branching ratio, vs. β_M, the metabolic scaling exponent), and final notation checks on the complex Womersley admittances: $Y_c \propto \sqrt{i\omega\rho Y_L}$, equivalently $Y_c \propto \sqrt{1 - F_{10}}$, and $M'_{10} = -(8i/\mathrm{Wo}^2)(1 - F_{10})$. We have also tightened the minimax presentation: Section 4 states the convex-envelope argument for existence and uniqueness, while Section 7 explicitly distinguishes the physiological cardiac duty cycle from the normalized adversarial coordinate $\eta \in [0, 1]$ and writes the endpoint maximum as the upper envelope of the two costs. Finally, several overstrong phrases (for example, "confirms," "necessity," and "zero free parameters") were softened and highlighted in magenta where they altered the submitted text; the quantitative predictions are unchanged.